

Adapters for Small Language Models

1 Introduction

Low-rank adaptation (LoRA) freezes the pretrained weights and injects small trainable matrices into every layer, which keeps the number of trainable parameters tiny. We study how this choice affects fine-tuning quality on small language models and report that the results hold across three model families.

A second paragraph starts with an indent and compares the method with full fine-tuning on a comparable budget of trainable parameters. The gap stays small on every task we tried, even when the rank is reduced to a single dimension for all attention layers.

2 Method

Each adapted layer adds a product of two thin matrices to its frozen weight. LoRA needs no extra inference latency because the product can be merged into the weight after training.